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Project Title:	Bioreactor Technology Transfer		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
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Space Biology Special Category:	None		
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Zip Code:	20892-1855	**Congressional District:**	8
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Task Description:	The NASA-NIH Center for Three-Dimensional Tissue Culture supports a number of scientists who work on projects related mainly to cell-cell interactions, cell differentiation, and the pathogenesis of infectious diseases. Many mainstream medical research studies have important implications for human space exploration and the Center helps promote the transfer of technology and interests between NASA and NIH. Intramural and extramural scientists launch pilot projects involving the RWV Bioreactor as extensions of their main research projects. Based on preliminary results, a research group can apply for a small grant proposal for a study involving the RWV. Additionally, many in-house projects are ongoing at the Center. Below are the current projects at the NASA/NIH Center: The effect of microgravity on the immune function of human lymphoid tissue (NICHD) Study of HIV-1 induced impairments (NICHD) Extracellular signals on differentiation of embryonic stem cells (NICHD) Study of the mechanism of malaria parasite release (NICHD) Culture conditions on membrane properties of Hab2 cells (NICHD) Development of ex vivo system to study Lyme disease (NCI) Role of extracellular matrix in cell differentiation (NIDCR) Characterization of RSV responses in human tonsil (CBER, FDA) Models of mucosal transmission of HIV infection (St. George's Hospital Medical School, London, England and Center for AIDS, UCLA) Culture of hematopoietic stem cells for gene therapy of hemoglobinopathies (Howard University College of Medicine, Washington, D.C.) Neurogenesis in a cell-hydrogel bioreactor system (Naval Research Laboratory, Washington D.C.) Replication of human intestinal parasite in intestinal cells (FDA, Laurel, MD) RWV models of human preneoplastic lesions in the lung (Dana-Farber Cancer Institute, Harvard Medical School, Boston, MA) The goal of the Center is to maintain a core facility at NIH for use of the NASA Rotating Wall Vessel (RWV) Bioreactor and related three-dimensional tissue culture systems for studying the basic mechanisms of normal and pathological intercellular interactions in multi-cellular systems, and how these interactions can be influenced by modeled microgravity. Investigators provide biological material or instruct the Center staff on experiment design, dissection and culture techniques. The investigator provides mentorship and is available for questions beyond staff expertise. The Center researchers prepare and culture the cells or tissues, and perform the primary analysis, including gathering data on tissue metabolism and tissue cytoarchitecture. The Center's staff also provides services such as immunocytochemistry, ELISA, confocal microscopy, electron microscopy, and flow cytometry. The primary investigator performs specialized analyses specific to their project. Successful pilot projects can then apply for funding, which keeps the staff of the Center available for new pilot projects, and promotes technology transfer by recruiting scientists to work on three-dimensional tissue culture full time. Because of our NASA funding, and because of the possibility that results obtained in suspension culture may relate to health problems of astronauts, or present unique opportunities for studying physiology in microgravity, there are possibilities of flight experiments for PIs of NASA/NIH projects. Currently we have flown one cellular experiment on the International Space Station (ISS), and we are currently running a fluid dynamics experiment on the ISS. The Center provides researchers an opportunity to develop new model systems for biomedical research and provides access to the RWV, which allows cellular growth, aggregation, and development of in vivo-like structures. This vessel allows the culture of cells under minimal shear forces in a well-oxygenated medium under conditions that mimic microgravity. RWVs and experienced staff are available for testing tissues, primary cell cultures, and cell lines under these conditions that facilitate cell-cell interactions and promote differentiation. NASA funding for equipment has enabled us to procure state-of-the-art microscopes to determine structure and function relationships between cells of developed tissues. The design of the Center allows investigators to assess potential benefits of the RWV for their specific aims without having to commit major resources, which greatly increases the number of investigators who will explore the RWV as a tissue culture opportunity and apply their knowledge and experience to space research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Since most of the studies conducted at the Center stem from mainstream medical research, the results obtained benefit not only space research, but the entire medical research community. In particular, the pathogenesis studies (HIV, malaria, Borrelia, RSV) and stem cell studies (embryonic, neural, hematopoietic) all have important Earth benefits.
	To support human exploration of space, it is essential that we assure that the highest standards of medical care be available to astronauts, even during space flights of extended duration. Towards that aim, it has been invaluable to the mission of space exploration to have a Center that promotes the exchange of technology and interests between NASA and NIH. In 2004, ten groups of intra- and extra-mural scientists worked at the NASA/NIH Center for Tissue Culture. These groups worked on thirteen various projects related to the effects of microgravity and modeled microgravity on both physiology and pathophysiology of human cells and tissues, as well as starting to develop the means for studying the behavior of stem cells in the NASA Rotating Wall Vessel Bioreactor (RWV). Following is a summary of results from ongoing projects at the Center: • We continue to investigate the effects of modeled and true microgravity on humoral immune responses of human lymphoid tissues and cells with the goal of understanding molecular and cellular mechanisms of compromised immunity in space and the development of countermeasures. Human lymphoid tissues and cells cultured in RWVs or on board ISS remain viable and metabolically active, however, they demonstrate impaired antibody and cytokine responses to recall antigens and polyclonal activators. However, if cells were challenged with activators prior to exposure to either modeled or true microgravity they maintain increased antibody and cytokine production. We hope to further implicate microgravity as a factor in immune dysfunction by ruling out the effects of physical suspension, cell shape, and adherence on cell membrane properties. Results of the ISS experiment prompted further studies, which are currently being performed by ISS astronauts to evaluate fluid/particle mixing in the cell culture chambers to determine optimal mixing and bubble removal procedures for microgravity experiments performed in the ISS tissue culture module. • Non-infectious HIV-1 virions have been found to inhibit immune response in human lymphoid tissue ex vivo, by production of a soluble factor by the tissue. We identified the cells producing this factor and the cells which are affected. We are now working to identify the factor, and to plan to use this system of immunodeficiency to study astronaut health issues such as viral reactivation, host-microbial interactions, and immunodeficiency/infection. • We have established three-dimensional cultures of rectal endoscopic biopsies and cervical and tonsillar tissues. We are standardizing the culture system for all tissues and using the same HIV-1 viral stocks to study effects of tissue environment on the replication of different viral phenotypes (R5, X4, R5/X4). This culture system will be used to study mechanisms of HIV-1 transmission across mucosal surfaces, test the ability of microbicides to prevent HIV-1 infection, and address host factors that influence the outcome of disease in immunocompromised tissues, which has implications for space-related diseases, such as the secondary immunodeficiency disease seen in astronauts. • We have developed ex vivo tonsil cultures to study functional responses of tonsil memory CD4 T cells and DC to

Task Progress:	respiratory syncytial virus (RSV). The ultimate goal is to apply this system to the design of vaccines and therapeutics that will attenuate RSV associated wheezing and RSV-enhanced allergic inflammation, and to apply the system to modeled microgravity to begin to address questions related to hypersensitivity in space. • We have demonstrated that human tonsil tissue cultured in RWV not only supports, but markedly enhances B. burgdorferi proliferation. We have also found that that there is a differential expression of Osp A and Osp C, indicating that the pathogenic spirochyte alters its gene expression as a function of the external milieu. We are now investigating B. burgdorferi gene and protein expression in the course of adaptation to human tissues and development of Lyme disease. Establishing a method of studying Borrelia adaptation will allow us monitor changes of adaptation in modeled microgravity which may help identify host-microbial interactions that are altered in microgravity and provide insight on countermeasures. • Using a biochemical approach, we studied the release mechanism of malaria parasites from red blood cells and have found that they are released by rupture and we are working to identify the release agents. We developed a large-scale method of malaria parasite replication in rotating bioreactors which has increased the yield of infected cells. By developing a detailed system of studying parasite infection and release from RBC, we can use this system to investigate changes in parasite production and pathogenicity in normal and modeled microgravity. • We have developed three-dimensional (3D) culture systems of embryonic stem (ES) cells and demonstrated that both soluble factors and the type of extracellular matrix (ECM) seem to be critical in directing differentiation and the formation of tissue-like structures. We are now investigating expression of adhesion molecules and differentiation markers in ES cells cultured 3D conditions and investigating in more detail the role of composition and topology of the ECM in ES cell growth factor responsiveness and the pattern of differentiation. Understanding the factors controlling ES differentiation is critical for many aspects of spaceflight as stem cell therapies are being proposed for many astronaut health issues. • A new study has been launched at the Center to test whether important cell membrane properties, such as the surface expression of HA and fusogenicity of HAb2 cells, are independent of cell culture conditions, in particular modeled microgravity. Alterations of these cell properties have important implications for susceptibility to pathogen infection, which is an area of great interest in spaceflight research and astronaut health. A number of projects supported by the Center were either completed, or support was no longer requested due to alternate funding. These include the following: • Rhesus monkey embryonic stem cells and human salivary gland cell line were used to study stem cell and acinar cell differentiation, respectively, in the RWV to study the influence of various types of extracellular matrix (ECM) on their differentiation. It was found that ECM induced differentiation not only occurs in RWVs but is often more extensive than conventional culture methods. • Developed a model of RWV-cultured intestinal cells for studying kinetics of Cyclospora infection, interactions between the parasites and infected cells, and the stages of life cycle occurring in the human gut. • Established culture of neural stem cells in engineered hydrogel scaffolds in RWVs to understand how modeled microgravity affects neural tissue generation from neural stem cells, and to develop an approach to engineering functional neuronal networks for potential use in neural tissue replacement therapy. • Assisted with establishing cultures of mouse hematopoietic stem cells in modeled microgravity toward the goal of stem therapy in space. • Airway biopsies cultured in RWVs were successfully used as a system for evaluating preneoplastic lesions in patients at risk for lung cancer, which is important since there is a lack of intermediate biomarkers for patients at risk for lung cancer.

Bibliography Type:	Description: (Last Updated: 10/09/2012)
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